Cooper Union Climate Action Plan

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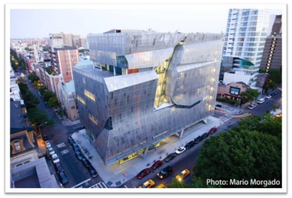
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Objectives

Cooper accepted the NYC Carbon Challenge aiming to reduce its carbon intensity by **50% by 2030** from 2014 peak emissions (**4752 → 2376 tCO2e**). Cooper also needs to comply with NYC **Local Law 97 (LL97)**, which establishes legally binding carbon limits for campus buildings by 2024, otherwise fines will be imposed. Cooper students are directly involved in climate action planning through real-time energy analysis, ventilation studies, and controls optimization, guided by faculty, staff and energy consultants. Since 2014, Cooper reduced its carbon footprint by **26%** but needs to achieve another **24%** to meet the NYC Carbon Challenge and maintain compliance with LL97.

Cooper Union



41 Cooper Square has two 500-ton chillers, natural gas hydronic heating, radiant floor and ceiling panels, dedicated outside air systems with air-side and water-side economizers, laboratory fume hoods and variable air volume systems.



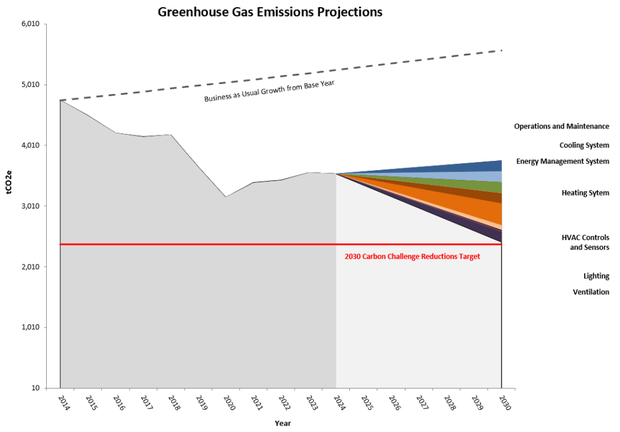
The Foundation Building has 300-ton chillers, natural gas condensing boilers, some upgraded air handlers and nine old McQuay hot and cold deck constant volume air handlers with pneumatic controls.

Methodology

Given that all of Coopers buildings are projected to incur LL97 penalties in the 2030s, Cooper Union is employing both industry expertise and student resources to meet its carbon reduction goal.

Industry-Side Energy Audit

Cooper students are performing an energy analysis of campus buildings under advisement of John Rundell and collaborating with facilities staff in establishing operational best practices and optimizing the building management system.



Operations and Maintenance: Repair and replacement of control valves and damper actuators, residence hall elevator replacement

Cooling: Upgrade chiller, repair/replacement of absorption chillers

Energy Management System: Air handling unit night (AHU) shutoff controls, variable frequency drive (VFD) and motor replacement

Heating: Optimize boilers, improve heat recovery, lab heat recovery, air source heat pump

HVAC Controls: AHU discharge air temperature optimization and static pressure resets, chilled and hot water differential pressure resets, optimized scheduling, chilled water and condenser water supply temperature optimization, vacancy sensor thermostats

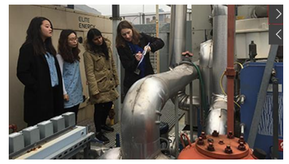
Lighting: LED lighting retrofit

Ventilation: Reduce outside air flow rates, laboratory demand controlled ventilation, HVAC demand response, repair air handlers

Student Involvement in Carbon Reduction

Cooper Union students have unique opportunities to work directly with industry professionals in achieving these Carbon Limits while gaining meaningful engineering knowledge and training for the energy industry. Current student opportunities and deliverables include:

1) An Energy Efficient Building Systems course (ME453), taught by John Rundell in Fall, which exposes students to energy efficient building systems and optimization.





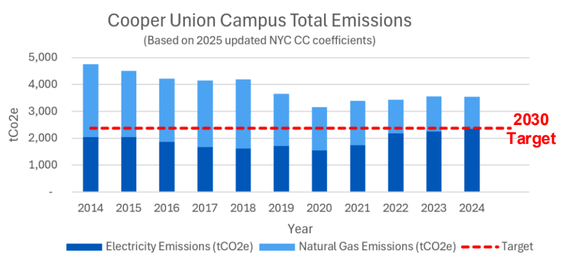
2) Apprenticeship-type site visits and inspection of mechanical rooms in the Foundation Building and Cooper Square.

3) Usage and exposure to Energy Tracking Software including Siemens Apogee, Energy Star Portfolio Manager, Veolia EMSYS and AVEVA Pi systems.



4) Forming undergraduate and Master's research projects centered on energy data analytics, building systems and controls optimization

Outcomes



As of 2024, Cooper has achieved 26% of the carbon reduction target. Since the cogeneration is no longer in use the ratio of natural gas to electricity emissions has decreased. However, there has been a plateau and slight increase since the year 2020.

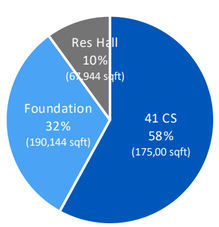
Climate action planning resulted in 14 Master's thesis, many VIP, Senior Capstone Design, and independent study projects.

14 publications, 6800 tons of CO_2 reduction, and ~ $3M in energy savings since 2014

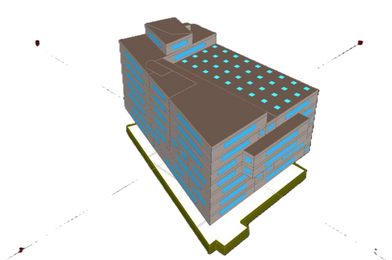
Foundation Building Energy Conservation Measures

- ECM 1A. Turn off fans from 12 AM -5 AM
- ECM 1B. Space temperature setbacks from 10 PM to 7 AM
- ECM 2. Operate Constant Volume as Variable Air Volume
- ECM 3. Convert Multizone Units to Variable Air Volume

Carbon Emission Contribution by Building



Foundation Building eQuest Model



41 CS Energy Conservation Measures

Every 41CS lab was audited to identify ventilation savings from shutting off fume and point hoods when not in use:

Total modeled reduction in CO2 was

200 tons

Total modeled energy savings were

310,000 kWh

Current 41CS AHUs supply air above design values. AHU-3, AHU-4, AHU-5, and AHU-6 were optimized with nighttime shutoff and zone-level airflow reduction. A baseline model was calibrated using trend data. The improved model used original design values and sequence of operations.

Total modeled energy savings were

685,000 kWh

Total modeled cost savings were

$200,000



